Week 03

The "Treat" Landscape

Natural Threats（自然威脅）: 地震、火災、水災等無法預測的事故

Unintentional Threats（無意威脅）: 不小心刪錯檔案、釣魚郵件

Internal Threats（內部）: 公司員工的惡意或報復行為

External Threats（外部）: 勒索軟體、DDoS、駭客師傅

Threat vs. Vulnerability vs. Risk

1. Threat	造成事件的因素	駭客集團
2. Vulnerability	系統或環境中存在的內在弱點	簡易密碼
3. Risk	威脅利用漏洞造成損失的機率與影響	伺服器被駭

Malware（Malicious software）

Common entry route: USB, browser, pirated software

Web-Based Distribution Techniques:

e.g. 駭入正版網站插入惡意程式碼、購買廣告讓假連結有更高的搜尋順位

Spear Phishing（針對式釣魚）

Drive-by-Download

e.g. 從網路下在程式、訪問執行 explolit 的 js 網站

The Midular Component of Malware

Cryptor	Protects malware from reverse engineering or analysis.
Downloader	A Trojan that downloads other malware from the internet.
Dropper	A Trojan that installs other malware files onto the system.
Exploit	Malicious code breaching security via software vulnerabilities.
Injector	Injects code into vulnerable running processes to hide or prevent removal.
Obfuscator	Conceals code and intended purpose, making detection hard.
Packer	Bundles files into a single executable via compression to bypass security detection (converts into unreadable format).
Payload	Software allowing control over a system after exploitation.
Malicious code	A command defining basic functionalities like stealing data and creating backdoors.

Trojans and Viruses

1. Trojans：負責偽裝並潛入系統
2. Viruses：負責感染系統、自我複製